

Use the logic gate symbols

9618_s24_qp_11_sc_1.b

QUESTION

(b) Draw a logic circuit for the logic expression:

$$X = \text{NOT} (\text{NOT } A \text{ AND } (\text{NOT } B \text{ XOR } C))$$



MARK SCHEME

1(b)	1 mark for NOT B XOR C 1 mark for NOT A and final AND plus NOT <pre>graph LR; A --> NA[NOT A]; B --> NB[NOT B]; C --> XOR[XOR]; NB --> XOR; XOR --> AND[AND]; NA --> AND; AND --> NX[NOT X]; NX --> X</pre>	2
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QUESTION

6 (a) Draw the logic circuit for this logic expression:

$$Z = (R \text{ XOR } S) \text{ AND } (\text{NOT } T \text{ NOR } P)$$



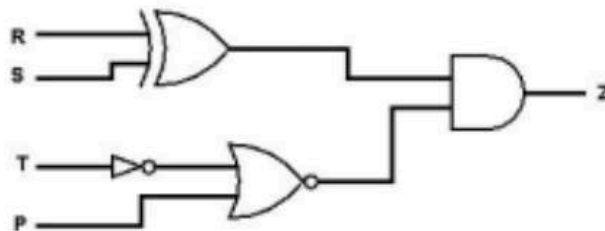
[2]

MARK SCHEME

6(a)

1 mark for correct XOR and AND gates, **with** correct inputs
1 mark for correct NOT and NOR gates **with** correct inputs

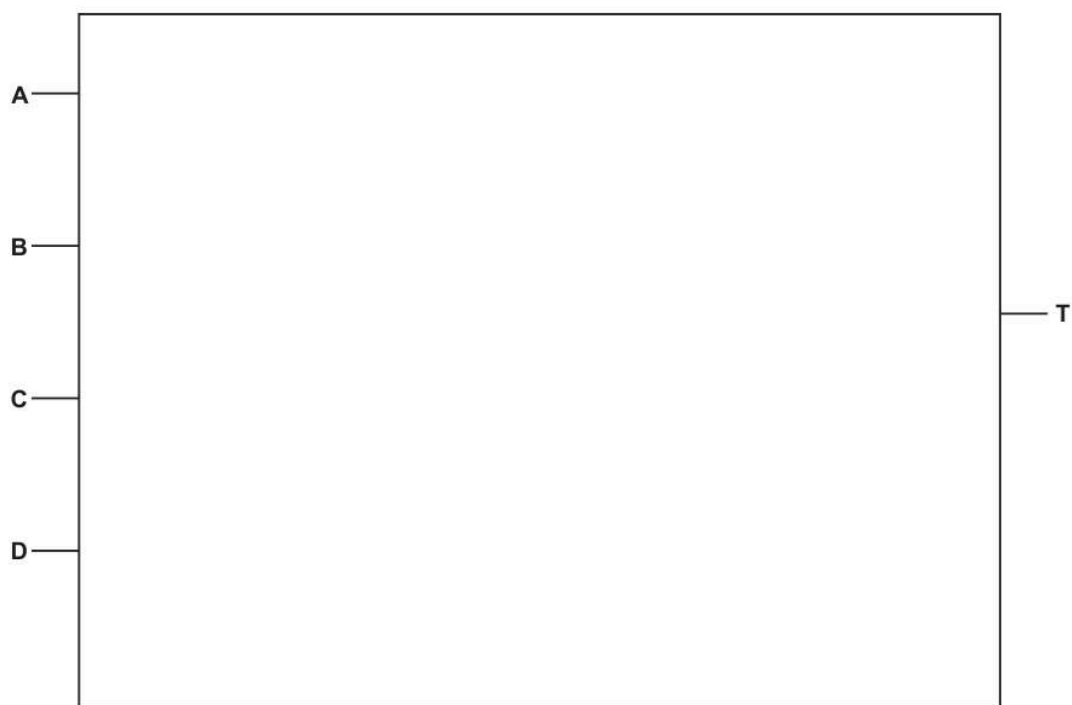
2



QUESTION

5 (a) Draw the logic circuit for this logic expression:

$$T = (\text{NOT } A \text{ OR } B) \text{ XOR } (C \text{ NAND } D)$$



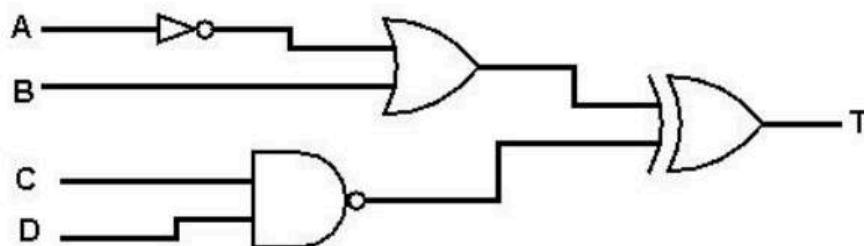
[2]

MARK SCHEME

5(a)

1 mark for 2 gates
2 marks for all 4 gates

2



QUESTION

(b) Draw a logic circuit for the following logic expression.

$$X = \text{NOT}(\text{NOT}(\text{A AND B}) \text{ AND } \text{C})$$



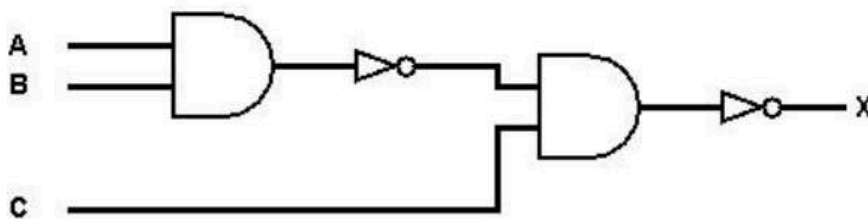
[2]

MARK SCHEME

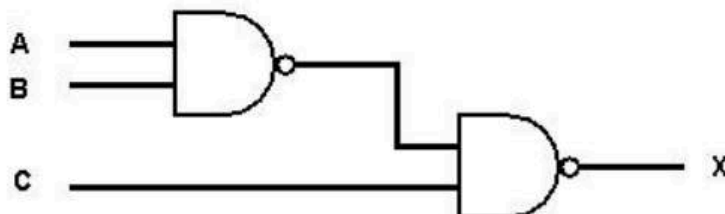
2(b)

1 mark for each correct bullet point

- NOT (A AND B) // A NAND B
- NOT the result AND C // the result NAND C



OR



2

QUESTION

- (c) Identify **one** logic gate **not** used in the given logic circuit. Draw the symbol for the logic gate **and** complete its truth table.

Logic gate:

Symbol:

Truth table:

A	B	Output
0	0	
0	1	
1	0	
1	1	

[3]

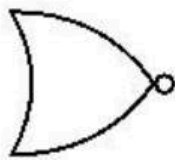
MARK SCHEME

3(c)

1 mark for gate, **1 mark** for matching symbol, **1 mark** for matching truth table

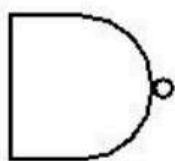
3

NOR



A	B	OUTPUT
0	0	1
0	1	0
1	0	0
1	1	0

NAND



A	B	OUTPUT
0	0	1
0	1	1
1	0	1
1	1	0

Understand and define the functions of: NOT, AND, OR, NAND, NOR and XOR (EOR) gates

9618_w25_qp_12_sc_9.b

QUESTION

(b) Consider the following logic expression:

$$X = \text{NOT } ((A \text{ NAND } B) \text{ XOR } (\text{NOT } A \text{ OR } \text{NOT } C))$$

A truth table for the logic expression is given:

Row number	A	B	C	X
1	0	0	0	1
2	0	0	1	0
3	0	1	0	1
4	0	1	1	0
5	1	0	0	1
6	1	0	1	0
7	1	1	0	1
8	1	1	1	1

There are **three** errors in the truth table.

Identify the **three** errors in the truth table by writing the row numbers with an incorrect output.

Error 1 Row number

Error 2 Row number

Error 3 Row number

[2]

MARK SCHEME

9(b)	1 mark for 1 or 2 row numbers correct, 2 marks all three correct Row 2 Row 4 Row 7	2
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QUESTION

1

(a)

Describe the operation of a 2-input XOR gate.

.....

..... [1]

MARK SCHEME

1(a)	1 mark for: Output is only 1 if one input is 1 and the other is 0 // Output is only 1 if both inputs are different // Output is only 0 if both inputs are the same	1
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QUESTION

1

(a)

Describe the operation of each of the following logic gates:

NAND

.....

NOR

.....

XOR

.....

OR

..... [4]

MARK SCHEME

1(a)	1 mark for each correct answer: <i>NAND</i> - The output is 0 when both inputs are 1, otherwise the output is 1 <i>NOR</i> - The output is 1 when both inputs are 0, otherwise the output is 0 <i>XOR</i> - The output is 1 when one of the inputs is 1 and the other input is 0, otherwise the output is 0 <i>OR</i> - The output is 0 when both inputs are 0, otherwise the output is 1	4
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9618_s23_qp_11_sc_5.b

QUESTION

(b) Describe the function of the NAND and NOR logic gates.

NAND

.....

NOR

.....

[2]

MARK SCHEME

5(b)	1 mark each <i>NAND</i> 0 is only output when both inputs are 1 // 1 is only output when none, or (either) one of the inputs is 1 <i>NOR</i> 1 is only output when both inputs are 0 // 0 is only output when (either) one or both inputs are 1	2
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QUESTION

8 Tick (✓) **one** box in each row to identify the logic gate that each statement describes.

Statement	AND	NAND	NOR	XOR	OR
The output is 1 only when both inputs are 1					
The output is 1 only when both inputs are different					
The output is 1 only when both inputs are 0					

[3]

MARK SCHEME

8

1 mark per correct row

3

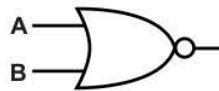
Statement	AND	NAND	NOR	XOR	OR
The output is 1 only when both inputs are 1	✓				
The output is 1 only when both inputs are different				✓	
The output is 1 only when both inputs are 0			✓		

Construct the truth table for each of the logic gates above

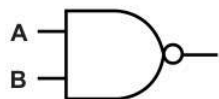
9618_w21_qp_12_sc_2.a

QUESTION

- 2 (a) Complete the truth table for each of the following two logic gates.



A	B	Output
0	0	
0	1	
1	0	
1	1	



A	B	Output
0	0	
0	1	
1	0	
1	1	

[2]

MARK SCHEME

2(a)	1 mark for each completely correct truth table NOR <table border="1"> <thead> <tr> <th>A</th><th>B</th><th>Output</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>0</td></tr> <tr><td>1</td><td>0</td><td>0</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </tbody> </table> NAND <table border="1"> <thead> <tr> <th>A</th><th>B</th><th>Output</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>0</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </tbody> </table>	A	B	Output	0	0	1	0	1	0	1	0	0	1	1	0	A	B	Output	0	0	1	0	1	1	1	0	1	1	1	0	2
A	B	Output																														
0	0	1																														
0	1	0																														
1	0	0																														
1	1	0																														
A	B	Output																														
0	0	1																														
0	1	1																														
1	0	1																														
1	1	0																														

9618_w21_qp_11_sc_3.c

QUESTION

- (c) Identify **one** logic gate **not** used in the given logic circuit. Draw the symbol for the logic gate **and** complete its truth table.

Logic gate:

Symbol:

Truth table:

A	B	Output
0	0	
0	1	
1	0	
1	1	

[3]

MARK SCHEME

3(c)	1 mark for gate, 1 mark for matching symbol, 1 mark for matching truth table NOR <table data-bbox="702 353 1177 685"> <thead> <tr> <th>A</th><th>B</th><th>OUTPUT</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>1</td></tr> <tr> <td>0</td><td>1</td><td>0</td></tr> <tr> <td>1</td><td>0</td><td>0</td></tr> <tr> <td>1</td><td>1</td><td>0</td></tr> </tbody> </table> NAND <table data-bbox="695 752 1171 1084"> <thead> <tr> <th>A</th><th>B</th><th>OUTPUT</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>1</td></tr> <tr> <td>0</td><td>1</td><td>1</td></tr> <tr> <td>1</td><td>0</td><td>1</td></tr> <tr> <td>1</td><td>1</td><td>0</td></tr> </tbody> </table>	A	B	OUTPUT	0	0	1	0	1	0	1	0	0	1	1	0	A	B	OUTPUT	0	0	1	0	1	1	1	0	1	1	1	0	3
A	B	OUTPUT																														
0	0	1																														
0	1	0																														
1	0	0																														
1	1	0																														
A	B	OUTPUT																														
0	0	1																														
0	1	1																														
1	0	1																														
1	1	0																														

Construct a logic circuit

9618_w25_qp_13_sc_3.b

QUESTION

(b) Draw a logic circuit for the logic expression:

$$X = \text{NOT } (((A \text{ AND } B) \text{ OR } C) \text{ AND } (C \text{ NOR } D))$$



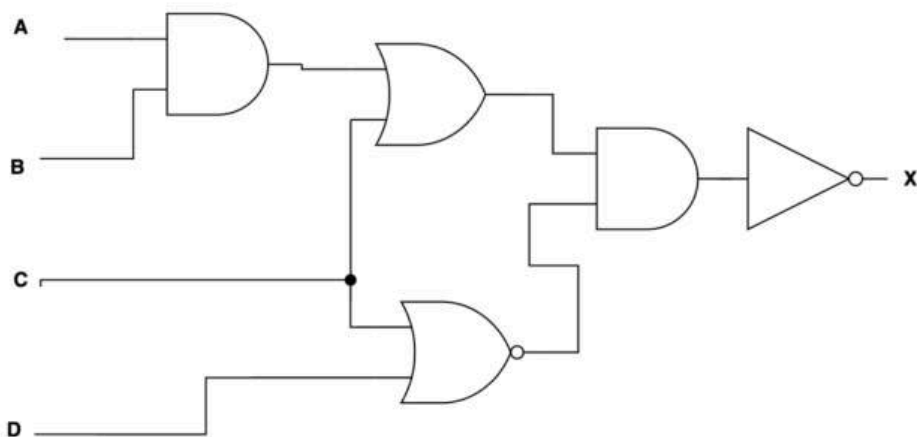
MARK SCHEME

3(b)

1 mark per bullet point, **max 2 marks**

2

- A AND B OR C
- C NOR D and final two gates



QUESTION

- 3 (a) Draw a logic circuit for the logic expression:

$$X = \text{NOT} ((A \text{ NAND } B) \text{ XOR } (\text{NOT } A \text{ OR } \text{NOT } C))$$



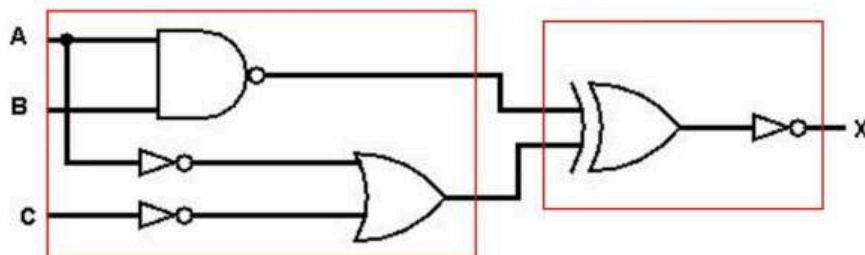
[2]

MARK SCHEME

3(a)

One mark per bullet point, **max 2** marks**2**

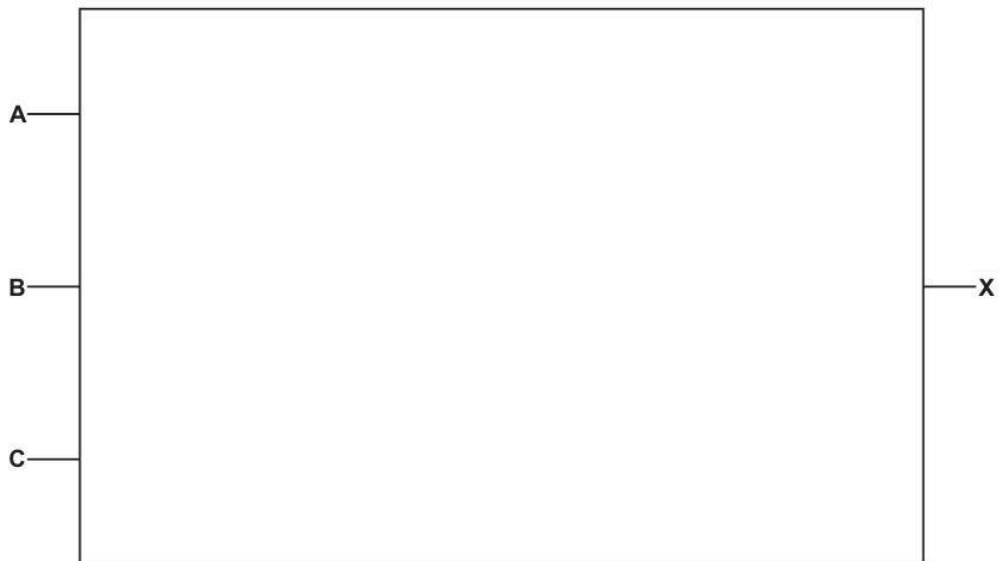
- (A NAND B) **and** (NOT A OR NOT C)
- XOR and NOT



QUESTION

(c) Draw the logic circuit for the logic expression:

$$X = ((\text{NOT } A \text{ AND } (B \text{ AND } C)) \text{ OR } (B \text{ NAND } C)) \text{ AND NOT } A$$



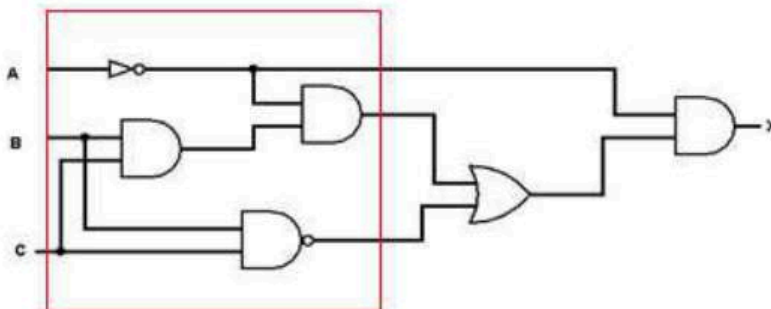
[2]

MARK SCHEME

1(c)

1 mark for NOT gate, 2 correct AND gates and the NAND gate
1 mark for OR gate and the final AND gate

2



QUESTION

(b) Draw the logic circuit for the logic expression:

$$W = P \text{ NAND } ((Q \text{ OR NOT } R) \text{ XOR } (P \text{ XOR } Q))$$



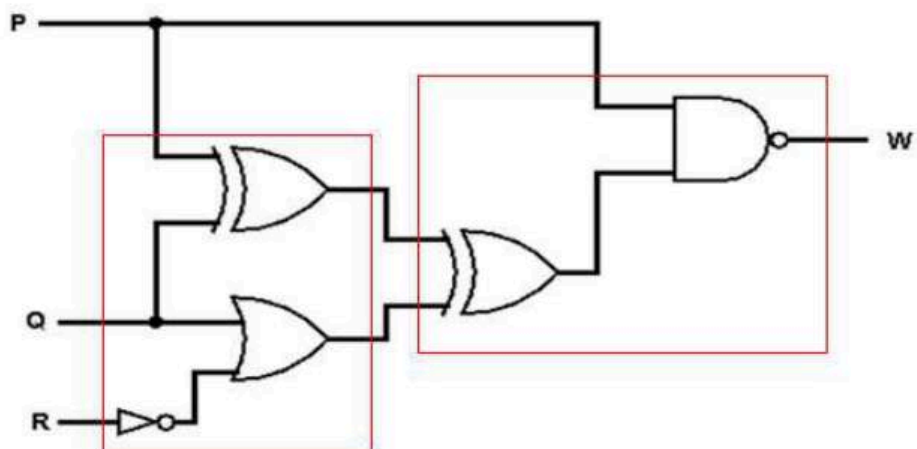
[2]

MARK SCHEME

1(b)

1 mark for (P XOR Q) and (Q OR NOT R)

1 mark for second XOR gate and final NAND gate with appropriate inputs

2

QUESTION

(b) Draw a logic circuit for this logic expression:

$$X = \text{NOT } ((A \text{ AND } B) \text{ OR } (C \text{ AND } D))$$

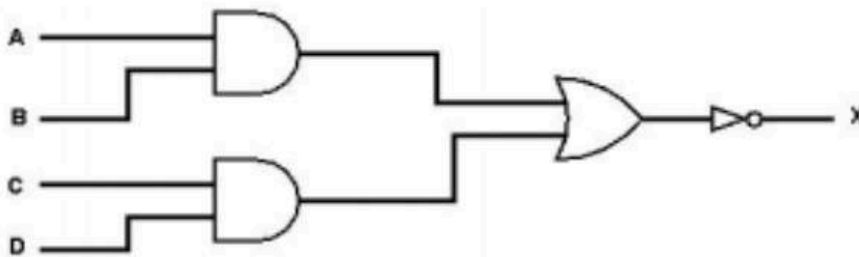


[2]

MARK SCHEME

1(b)

1 mark for both AND gates with correct inputs
1 mark for correct OR and NOT gates with correct inputs and no superfluous gates:

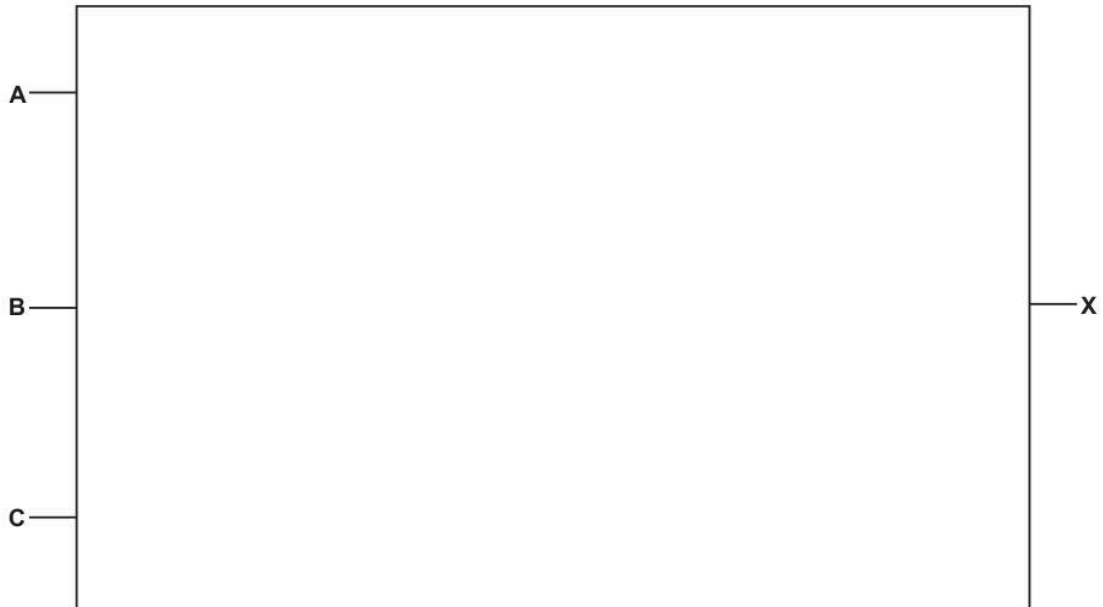


2

QUESTION

(b) Draw a logic circuit for the logic expression:

$$X = \text{NOT} (\text{NOT } A \text{ AND } (\text{NOT } B \text{ XOR } C))$$

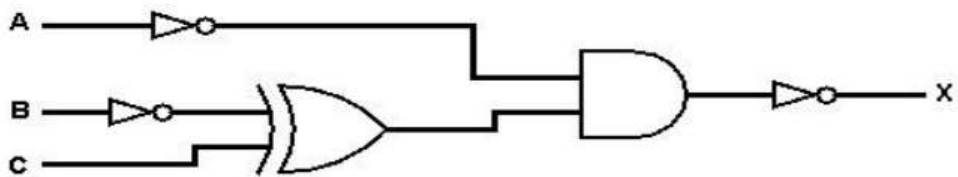


[2]

MARK SCHEME

1(b)

1 mark for NOT B XOR C
1 mark for NOT A and final AND plus NOT



2

QUESTION

(b) Draw a logic circuit for the logic expression:

$$Y = ((P \text{ AND } Q) \text{ XOR } ((\text{NOT } Q) \text{ OR } R)) \text{ AND NOT } P$$



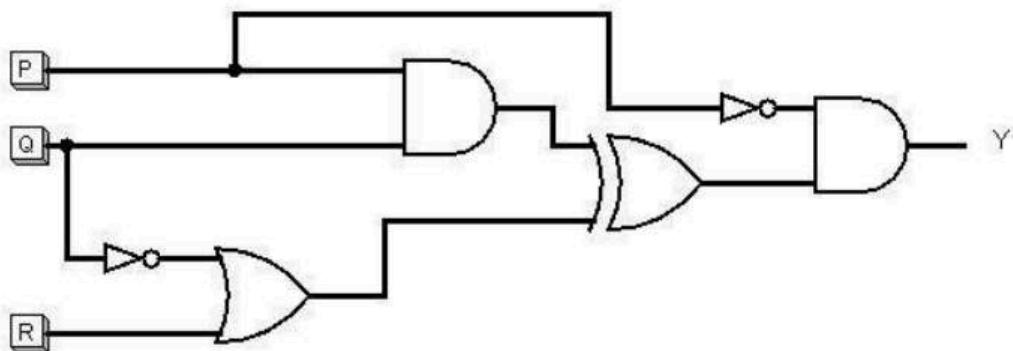
[2]

MARK SCHEME

4(b)

1 mark for **P AND Q** and **NOT Q OR R**
1 mark for **NOT gate** and **AND gate** and **XOR gate**

2



QUESTION

(b) Draw a logic circuit for the logic expression:

$$X = (A \text{ AND NOT } (B \text{ OR } C)) \text{ AND } (B \text{ NOR } C)$$



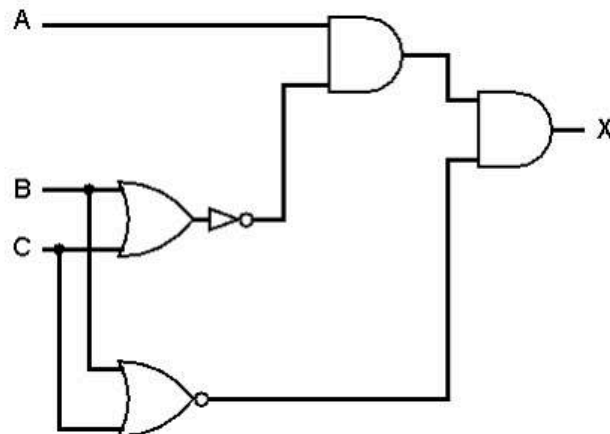
[2]

MARK SCHEME

4(b)

1 mark for both AND gates
1 mark for NOT gate and OR gate and NOR gate

2



QUESTION

6 (a) Draw the logic circuit for this logic expression:

$$Z = (R \text{ XOR } S) \text{ AND } (\text{NOT } T \text{ NOR } P)$$



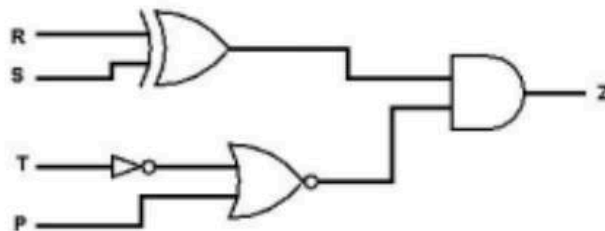
[2]

MARK SCHEME

6(a)

1 mark for correct XOR and AND gates, **with** correct inputs
1 mark for correct NOT and NOR gates **with** correct inputs

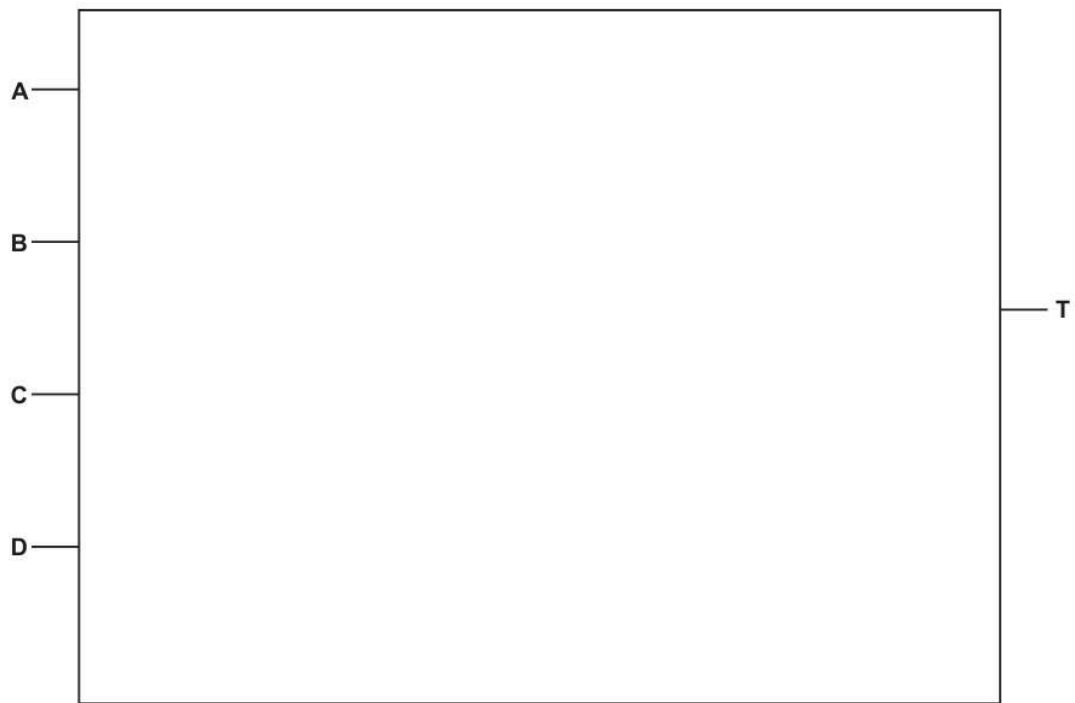
2



QUESTION

- 5 (a) Draw the logic circuit for this logic expression:

$$T = (\text{NOT } A \text{ OR } B) \text{ XOR } (C \text{ NAND } D)$$



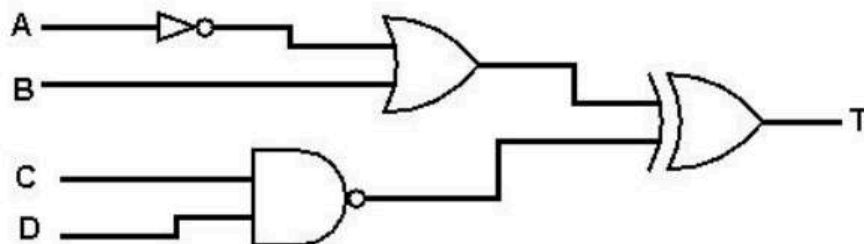
[2]

MARK SCHEME

5(a)

1 mark for 2 gates
2 marks for all 4 gates

2



QUESTION

- 5 (a) Draw a logic circuit for the logic expression:

$$X = \text{NOT} ((\text{NOT} (A \text{ AND } B)) \text{ OR } (\text{NOT} (B \text{ AND } C)))$$



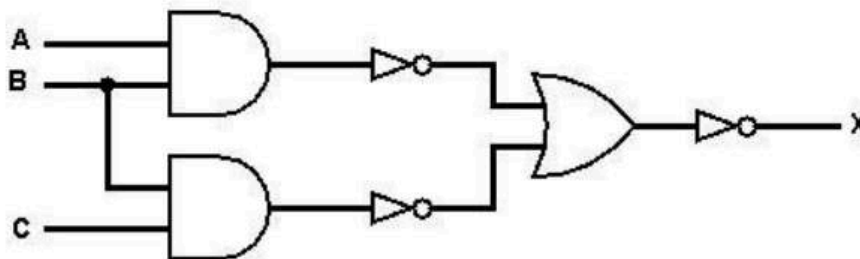
[3]

MARK SCHEME

5(a)

1 mark for each bullet point:

- NOT (A AND B)
- NOT (B AND C)
- NOT(NOT(A AND B) OR NOT(B AND C))

**3**

QUESTION

- 3 (a) A greenhouse has an automatic window.

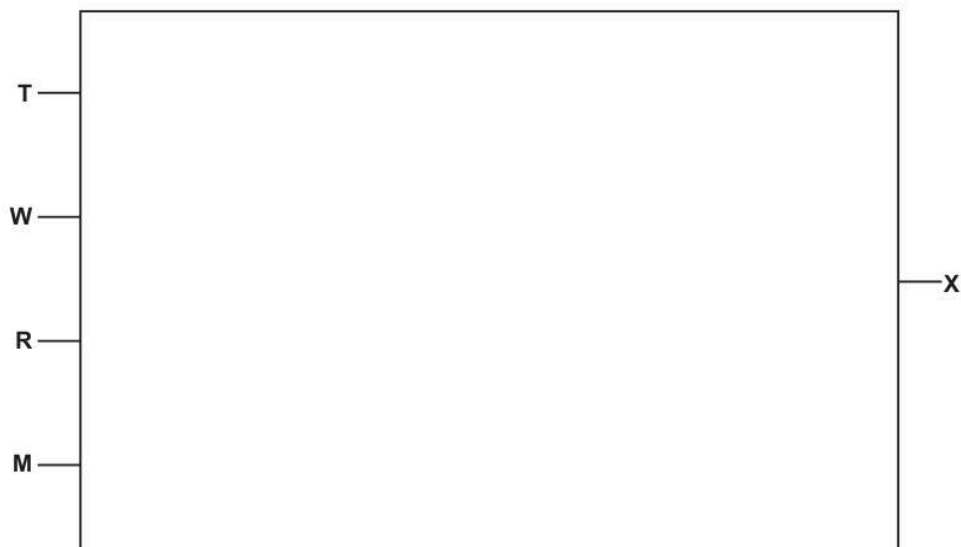
The window (**X**) operates according to the following criteria:

Parameter	Description of parameter	Binary value	Condition
T	Temperature	1	Too high
		0	Acceptable
W	Wind speed	1	Too high
		0	Acceptable
R	Rain	1	Detected
		0	Not detected
M	Manual override	1	On
		0	Off

The window opens (**X** = 1) if:

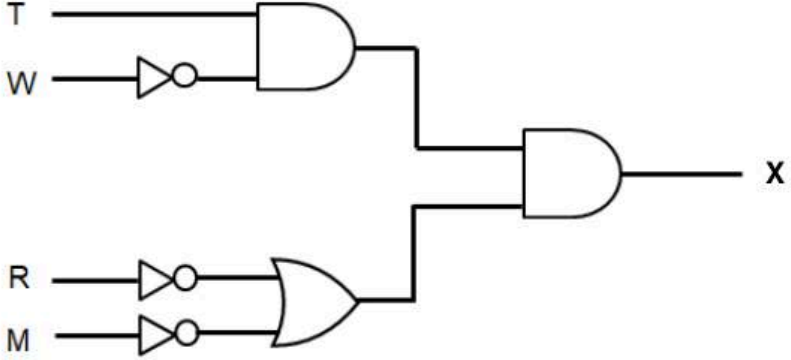
- the temperature is too high **and** the wind speed is acceptable
- **and**
- rain is not detected, **or** the manual override is off.

Draw a logic circuit to represent the operation of the window.



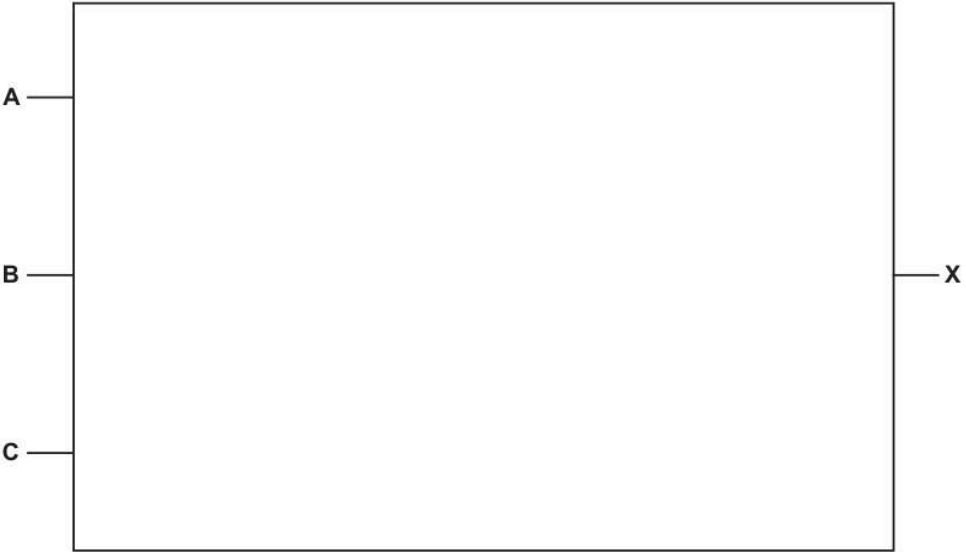
[3]

MARK SCHEME

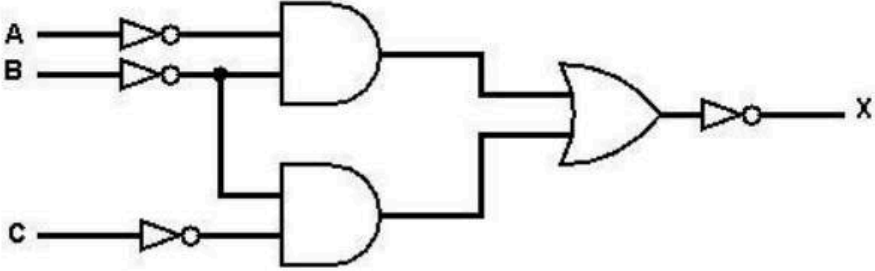
3(a)	1 mark for T AND NOT W 1 mark for NOT R OR NOT M 1 mark for final AND 	3
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9618_w22_qp_11_sc_3.a

QUESTION

3 (a) Draw a logic circuit for the logic expression: $X = \text{NOT } ((\text{NOT } A \text{ AND NOT } B) \text{ OR } (\text{NOT } B \text{ AND NOT } C))$  [2]
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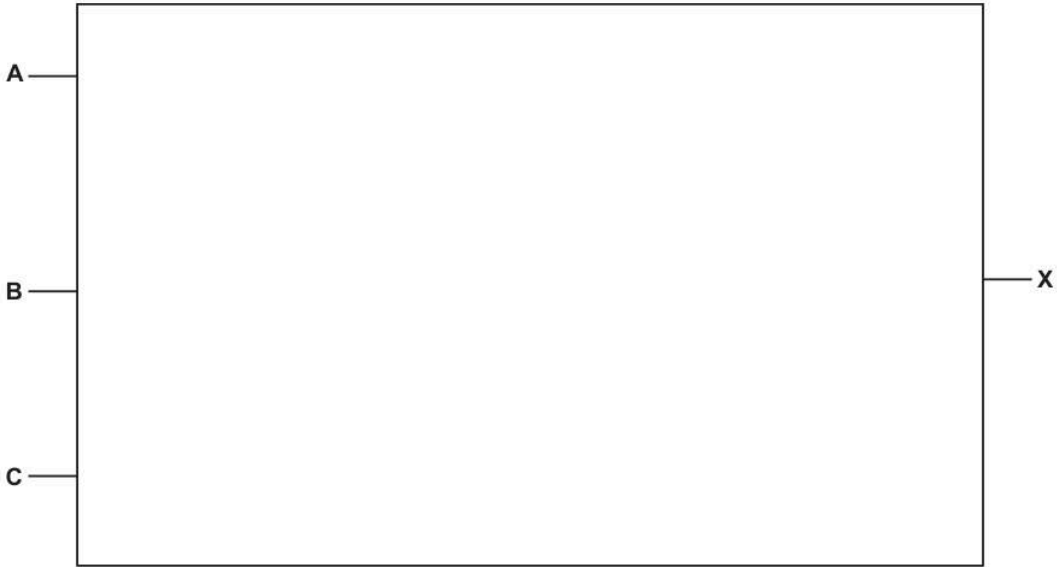
MARK SCHEME

3(a)	1 mark for each bullet point: • NOT A AND NOT B and NOT B AND NOT C // A NOR B and B NOR C • final OR and NOT gates (with correct inputs) // NOR gate (with correct inputs) 	2
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9618_w21_qp_12_sc_2.b

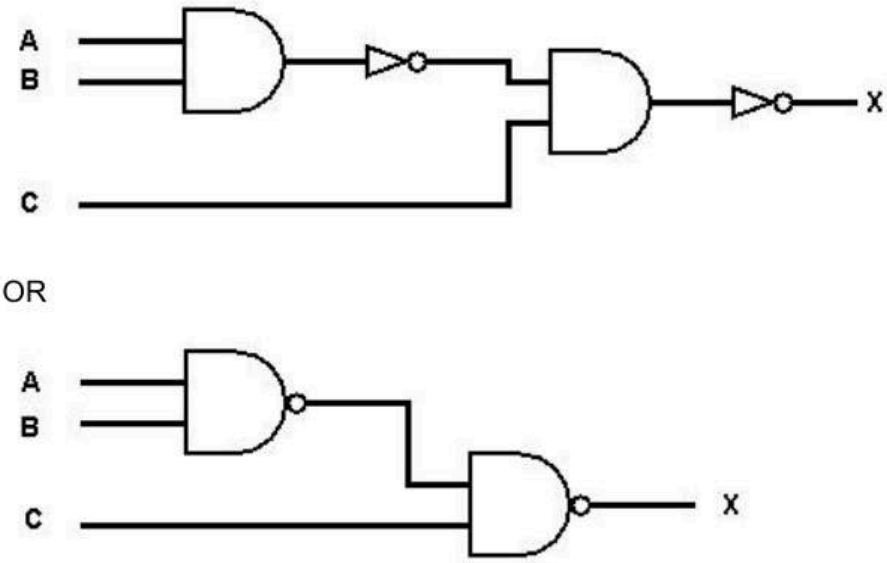
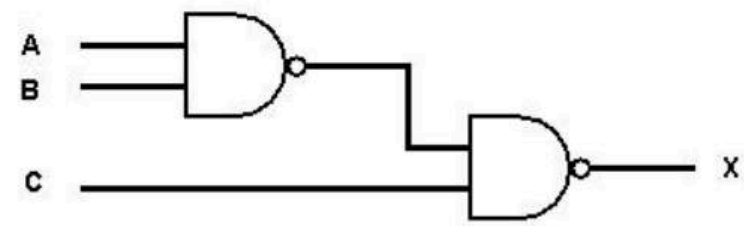
QUESTION

(b) Draw a logic circuit for the following logic expression.

$$X = \text{NOT}(\text{NOT}(A \text{ AND } B) \text{ AND } C)$$


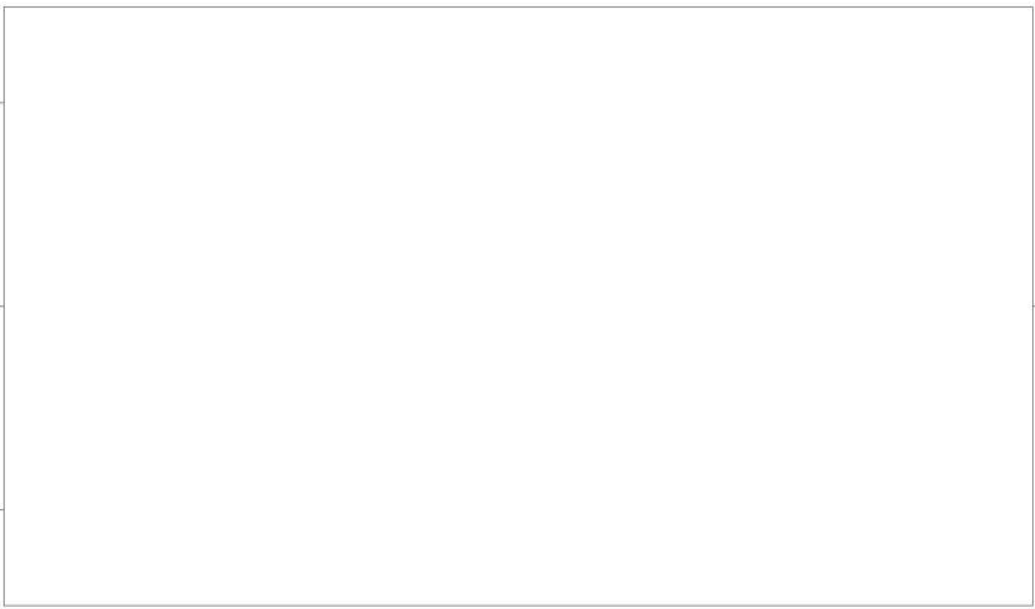
[2]

MARK SCHEME

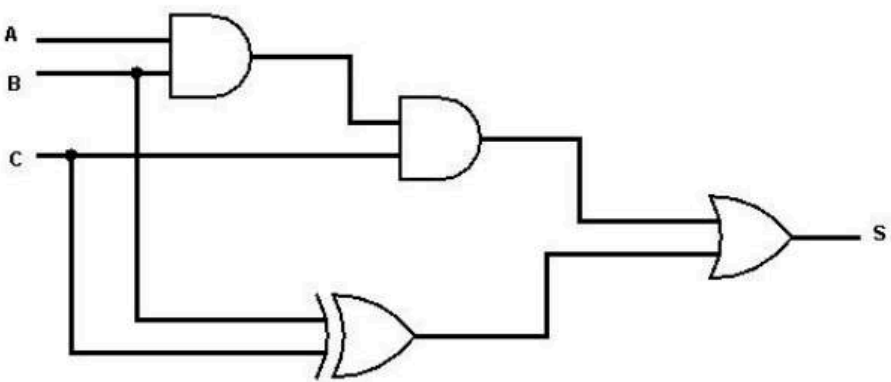
2(b)	1 mark for each correct bullet point • NOT (A AND B) // A NAND B • NOT the result AND C // the result NAND C  OR 	2
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9618_s21_qp_12_sc_3.a

QUESTION

3	A logic expression is given: $S = (A \text{ AND } B \text{ AND } C) \text{ OR } (B \text{ XOR } C)$ (a) Draw the logic circuit for the given expression. 	[4]
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MARK SCHEME

3(a)	1 mark for each correct gate, with correct inputs  <pre>graph LR; A --- AND1[AND]; B --- AND1; AND1 --- AND2[AND]; C --- AND2; C --- OR1[OR]; AND2 --- OR1; OR1 --- S</pre>	4
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Construct a truth table

9618_w25_qp_13_sc_3.a

QUESTION

- 3 (a) Complete the truth table for the following logic expression:

$$X = (A \text{ NOR } B) \text{ NAND } (C \text{ XOR } B)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

MARK SCHEME

3(a)

1 mark per shaded area, max 2 marks

2

A	B	C		X
0	0	0		1
0	0	1		0
0	1	0		1
0	1	1		1
1	0	0		1
1	0	1		1
1	1	0		1
1	1	1		1

QUESTION

(b) Consider the following logic expression:

$$X = \text{NOT } ((A \text{ NAND } B) \text{ XOR } (\text{NOT } A \text{ OR } \text{NOT } C))$$

A truth table for the logic expression is given:

Row number	A	B	C	X
1	0	0	0	1
2	0	0	1	0
3	0	1	0	1
4	0	1	1	0
5	1	0	0	1
6	1	0	1	0
7	1	1	0	1
8	1	1	1	1

There are **three** errors in the truth table.

Identify the **three** errors in the truth table by writing the row numbers with an incorrect output.

Error 1 Row number

Error 2 Row number

Error 3 Row number

[2]

MARK SCHEME

9(b)	1 mark for 1 or 2 row numbers correct, 2 marks all three correct Row 2 Row 4 Row 7	2
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QUESTION

(b) Complete the truth table for the logic expression:

$$X = (A \text{ OR } B) \text{ XOR } (B \text{ OR } C) \text{ XOR } (\text{NOT } A \text{ NAND } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

MARK SCHEME

4(b)

1 mark for 1st 4 rows
1 mark for 2nd 4 rows

A	B	C	Working space	X
0	0	0		1
0	0	1		1
0	1	0		1
0	1	1		0
1	0	0		0
1	0	1		1
1	1	0		1
1	1	1		1

2

QUESTION

(b) Complete the truth table for the logic expression:

$$X = \text{NOT } (A \text{ AND } (B \text{ XOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

MARK SCHEME

4(b)

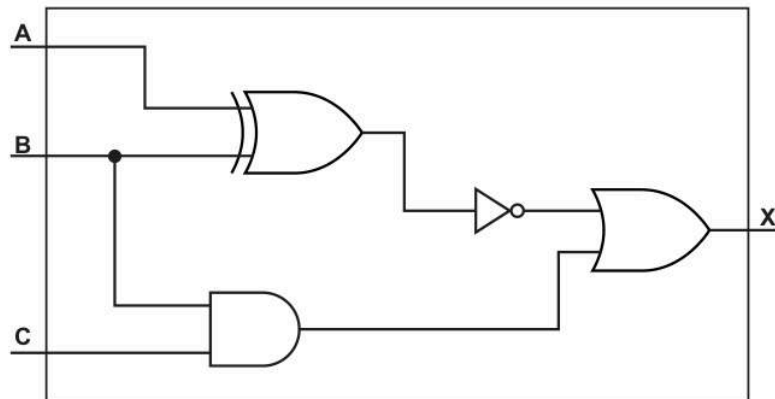
1 mark for first four rows, 1 mark for second four rows

2

A	B	C	Working space	X
0	0	0		1
0	0	1		1
0	1	0		1
0	1	1		1
1	0	0		1
1	0	1		0
1	1	0		0
1	1	1		1

QUESTION

(b) Complete the truth table for the following logic circuit.



A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

MARK SCHEME

1(b)

1 mark for first 4 rows, 1 mark for second 4 rows

2

A	B	C	Working space	X
0	0	0		1
0	0	1		1
0	1	0		0
0	1	1		1
1	0	0		0
1	0	1		0
1	1	0		1
1	1	1		1

9618_w24_qp_12_sc_1.a

QUESTION

- 1 (a) Complete the truth table for the logic expression:

$$X = (A \text{ XOR } B) \text{ NAND } (A \text{ AND } (B \text{ XOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

MARK SCHEME

1(a)

1 mark for each shaded part.

2

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

QUESTION

12

- 6 Each of the following truth tables has three inputs (**A**, **B** and **C**) and one output (**X**).

Draw **one** line to match each truth table with its logic expression.

Truth table

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Logic expression

NOT (A XOR B) AND C

(A OR C) AND NOT B

A NAND B NAND C

(A NAND B) OR C

NOT (A AND B AND C)

[3]

MARK SCHEME

6

1 mark for each correct line:

3

Truth table

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Logic expression

NOT (A XOR B) AND C

(A OR C) AND NOT B

A NAND B NAND C

(A NAND B) OR C

NOT (A AND B AND C)

QUESTION

- 4 (a) Complete the truth table for the logic expression:

$$Y = ((P \text{ AND } Q) \text{ XOR } ((\text{NOT } Q) \text{ OR } R)) \text{ AND NOT } P$$

P	Q	R	Working space	Y
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

MARK SCHEME

4(a)	1 mark for each set of highlighted rows	2																																				
<table><tr><th>P</th><th>Q</th><th>R</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table>			P	Q	R	Y	0	0	0	1	0	0	1	1	0	1	0	0	0	1	1	1	1	0	0	0	1	0	1	0	1	1	0	0	1	1	1	0
P	Q	R	Y																																			
0	0	0	1																																			
0	0	1	1																																			
0	1	0	0																																			
0	1	1	1																																			
1	0	0	0																																			
1	0	1	0																																			
1	1	0	0																																			
1	1	1	0																																			

QUESTION

(b) Complete the truth table for the logic expression:

$$X = A \text{ XOR } (B \text{ AND } (A \text{ NAND } B) \text{ AND NOT } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

MARK SCHEME

4(b)

1 mark for each set of rows (shaded)

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

2

QUESTION

- 4 (a) Complete the truth table for the logic expression:

$$X = \text{NOT } (A \text{ NAND } B) \text{ XOR } (\text{NOT } B \text{ AND } (B \text{ NOR } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

MARK SCHEME

4(a)

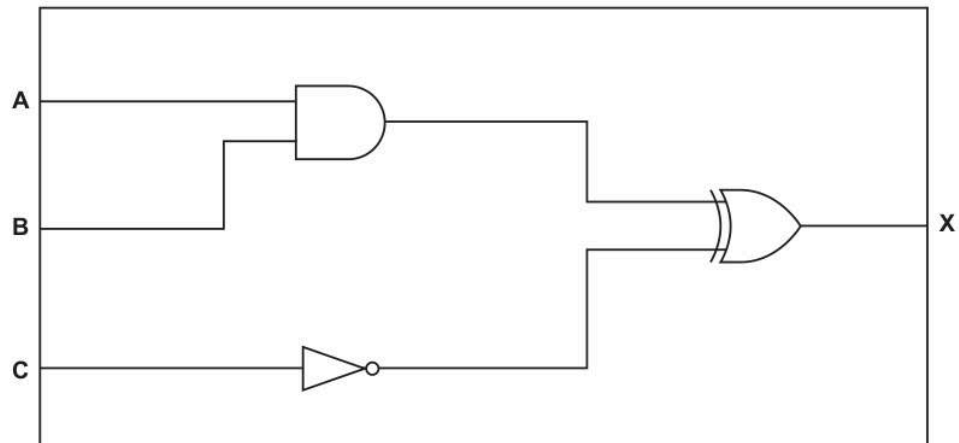
1 mark for each set of 4 rows (shaded)

2

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

QUESTION

(b) Complete the truth table for this logic circuit:



A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

MARK SCHEME

1(b)	1 mark for first 4 rows correct; 1 mark for second 4 rows correct	2																																				
<table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>			A	B	C	X	0	0	0	1	0	0	1	0	0	1	0	1	0	1	1	0	1	0	0	1	1	0	1	0	1	1	0	0	1	1	1	1
A	B	C	X																																			
0	0	0	1																																			
0	0	1	0																																			
0	1	0	1																																			
0	1	1	0																																			
1	0	0	1																																			
1	0	1	0																																			
1	1	0	0																																			
1	1	1	1																																			

9618_s23_qp_12_sc_6.b

QUESTION

(b) Complete the truth table for this logic expression:

$$Z = (\text{NOT } P \text{ OR } Q) \text{ XOR } (R \text{ NOR } Q)$$

P	Q	R	Working space	Z
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

MARK SCHEME

6(b)

1 mark for first 4 rows
1 mark for last 4 rows

2

P	Q	R	Working space	Z
0	0	0		0
0	0	1		1
0	1	0		1
0	1	1		1
1	0	0		1
1	0	1		0
1	1	0		1
1	1	1		1

9618_w22_qp_13_sc_5.b

QUESTION

(b) Complete the truth table for the logic expression:

$$Y = (\text{NOT } P \text{ AND } Q) \text{ OR } (Q \text{ AND NOT } R)$$

P	Q	R	Working space	Y
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

MARK SCHEME

5(b)

1 mark for each set of highlighted rows.

2

P	Q	R	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

9618_w22_qp_12_sc_3.b

QUESTION

(b) Complete the truth table for the logic expression:

$$X = \text{NOT } (A \text{ OR } B \text{ OR } C) \text{ AND } (B \text{ NOR } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

MARK SCHEME

3(b)

1 mark for each set of rows as highlighted:

2

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

9618_w22_qp_11_sc_3.b

QUESTION

(b) Complete the truth table for the logic expression:

$$X = \text{NOT } ((\text{NOT } A \text{ AND NOT } B) \text{ OR } (\text{NOT } B \text{ AND NOT } C))$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

MARK SCHEME

3(b)

1 mark for each set of rows as highlighted:

2

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

9618_w21_qp_11_sc_3.b

QUESTION

(b) Complete the truth table for the given logic circuit.

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

MARK SCHEME

3(b)	1 mark for each set of 4 rows (shaded)	2																																													
<table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr></table>			A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		0	0	1	1		1	1	0	0		1	1	0	1		1	1	1	0		1	1	1	1		0
A	B	C	Working space	X																																											
0	0	0		1																																											
0	0	1		1																																											
0	1	0		0																																											
0	1	1		1																																											
1	0	0		1																																											
1	0	1		1																																											
1	1	0		1																																											
1	1	1		0																																											

9618_s21_qp_12_sc_3.b

QUESTION

(b) Complete the truth table for the logic expression:

$$S = (A \text{ AND } B \text{ AND } C) \text{ OR } (B \text{ XOR } C)$$

A	B	C	Working space	S
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

MARK SCHEME

3(b)

1 mark for each half (shaded)

2

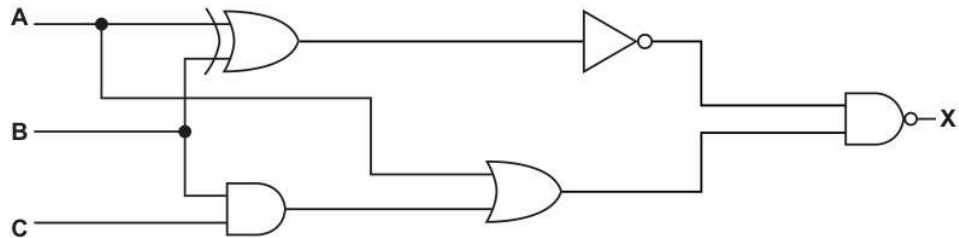
A	B	C	Working space	S
0	0	0		0
0	0	1		1
0	1	0		1
0	1	1		0
1	0	0		0
1	0	1		1
1	1	0		1
1	1	1		1

Construct a logic expression

9618_w25_qp_12_sc_9.a

QUESTION

9 (a) Consider the following logic circuit:



Write the logic expression for the logic circuit. Do **not** simplify the expression.

X = [2]

MARK SCHEME

9(a)	1 mark per bullet point, max 2 marks - NOT (A XOR B) - NAND ((B AND C) OR A) X = (NOT (A XOR B)) NAND ((B AND C) OR A) // X = ((B AND C) OR A) NAND (NOT (A XOR B))	2
------	---	---

QUESTION

(b) Write the logic expression for the following truth table.

R	S	T	Q
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

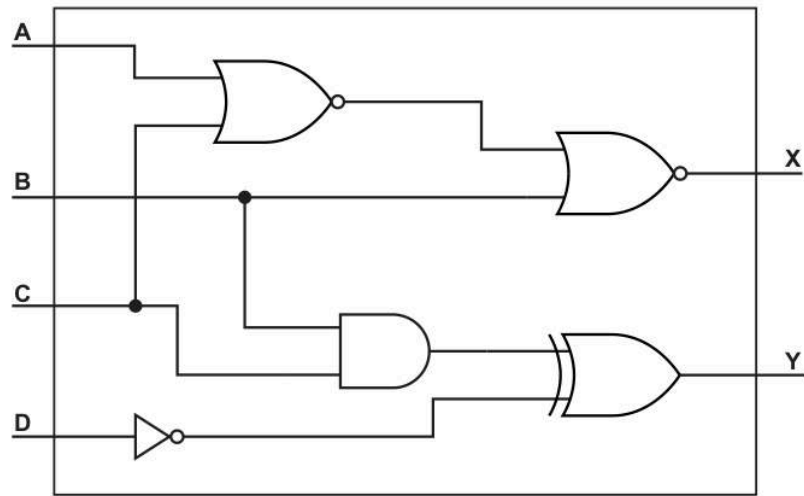
Q = [2]

MARK SCHEME

3(b)	1 mark per bullet point, max 2 marks - One correct term: (R AND S AND NOT T) // (NOT R AND NOT S AND T) - Second correct term plus the OR in correct place Answer: Q = (R AND S AND NOT T) OR (NOT R AND NOT S AND T)	2
------	---	----------

QUESTION

- 4 (a) Write the logic expressions for the following logic circuit.



X =

Y =

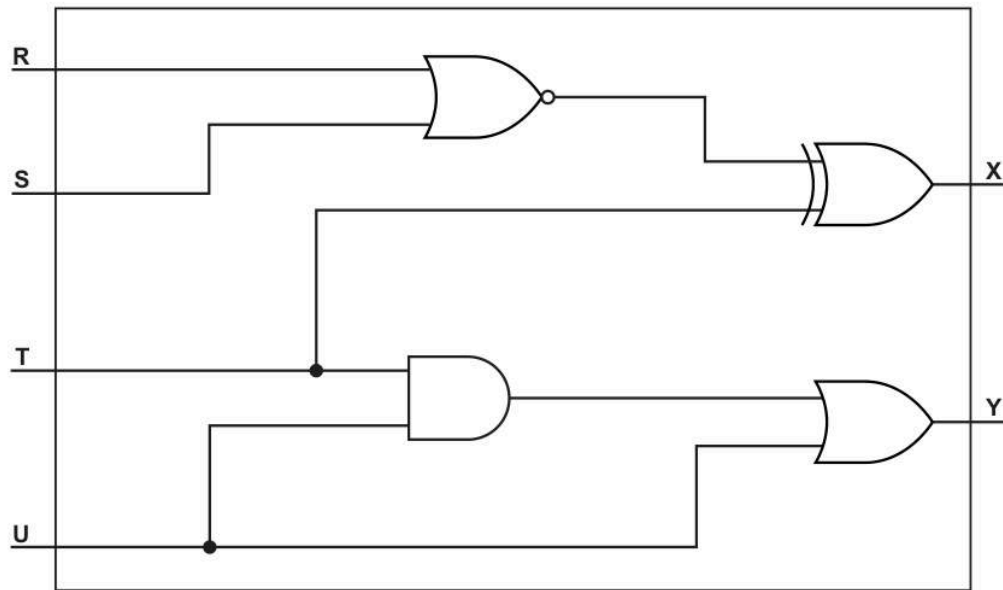
[2]

MARK SCHEME

4(a)	1 mark for each correct statement $X = B \text{ NOR } (A \text{ NOR } C)$ $Y = (B \text{ AND } C) \text{ XOR NOT } D$	2
------	---	----------

QUESTION

- 4 (a) Write the logic expressions for the following logic circuit.



X =

Y =

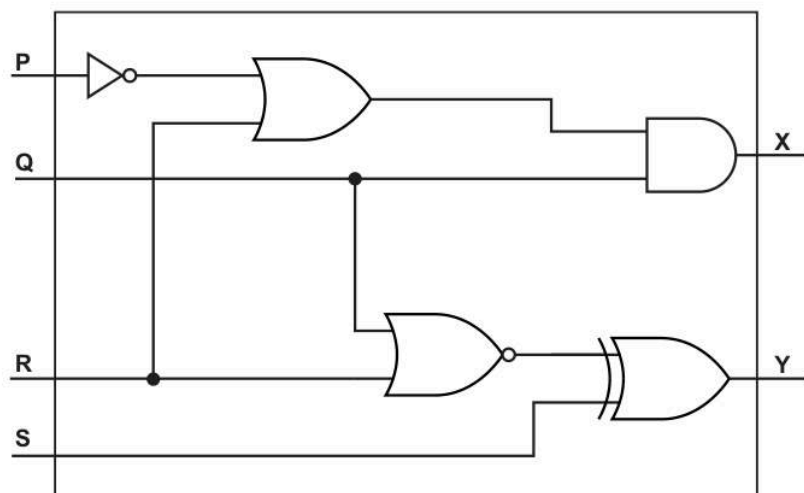
[2]

MARK SCHEME

4(a)	1 mark each: • $X = (R \text{ NOR } S) \text{ XOR } T$ • $Y = (T \text{ AND } U) \text{ OR } U$	2
------	---	----------

QUESTION

- 1 (a) Write the logic expressions for the following logic circuit.



X =

Y =

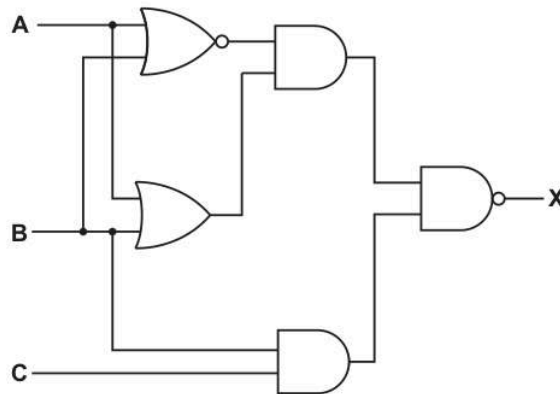
[2]

MARK SCHEME

1(a)	1 mark each: $X = Q \text{ AND } (\text{NOT } P \text{ OR } R)$ $Y = S \text{ XOR } (Q \text{ NOR } R)$	2
------	--	----------

QUESTION

(b) Write the logic expression for the following logic circuit.



X =
 [2]

MARK SCHEME

1(b)	1 mark each: • ((A NOR B) AND (A OR B)) • NAND (B AND C) X = ((A NOR B) AND (A OR B)) NAND (B AND C)	2
------	---	----------

QUESTION

(b) The floodlight (**X**) and audio alarm (**Y**) operate according to the following criteria:

Parameter	Description of parameter	Binary value	Condition
A	external daylight level	1	Low
		0	High
B	front door	1	Open
		0	Closed
C	person is within 2 m	1	Detected
		0	Not detected
D	back door	1	Open
		0	Closed
E	security system	1	Switched on
		0	Switched off

The floodlight turns on (**X = 1**) if:

- the security system is switched on
and
- the external daylight level is low
and
- a person is detected within 2 m.

The audio alarm turns on (**Y = 1**) if:

- the security system is switched on
and
- one **or** more doors are open, **or** a person is detected within 2 m.

Write logic expressions for the security system.

X =

Y =
 [2]

MARK SCHEME

5(b)	1 mark for each correct expression X = E AND A AND C Y = E AND (B OR C OR D)	2
------	---	----------

QUESTION

12

- 6 Each of the following truth tables has three inputs (**A**, **B** and **C**) and one output (**X**).

Draw **one** line to match each truth table with its logic expression.

Truth table

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Logic expression

NOT (A XOR B) AND C

(A OR C) AND NOT B

A NAND B NAND C

(A NAND B) OR C

NOT (A AND B AND C)

[3]

MARK SCHEME

6

1 mark for each correct line:

3

Truth table

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Logic expression

NOT (A XOR B) AND C

(A OR C) AND NOT B

A NAND B NAND C

(A NAND B) OR C

NOT (A AND B AND C)

QUESTION

- 1 (a) Tick (✓) **one** box to identify the correct logic statement for this truth table.

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

- ☐ NOT (A AND B AND C)
- ☐ (A XOR B) NOR C
- ☐ (A OR B OR C) NOR C
- ☐ NOT A AND NOT B AND NOT C

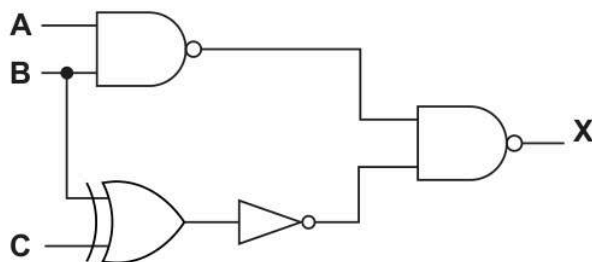
[1]

MARK SCHEME

1(a)	1 mark for: (A XOR B) NOR C	1
------	--------------------------------	---

QUESTION

- 4 (a) Write the Boolean expression that corresponds to the following logic circuit.



.....

..... [3]

MARK SCHEME

4(a)	1 mark for each bullet point - A NAND B - NOT(B XOR C) - Final NAND $X = A \text{ NAND } B \text{ NAND } (\text{NOT}(B \text{ XOR } C))$	3
------	---	----------

9618_s23_qp_13_sc_1.a

QUESTION

- 1 (a) Write the logic expression for this truth table:

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

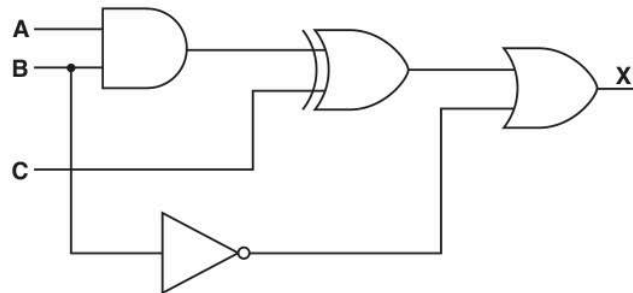
..... [1]

MARK SCHEME

1(a)	NOT B	1
------	-------	----------

QUESTION

3 A logic circuit is shown:



(a) Write the logic expression for the logic circuit.

.....
 [3]

MARK SCHEME

3(a)	1 mark per bullet point • A AND B ... • ... XOR C ... • ... OR NOT B ((A AND B) XOR C) OR NOT B	3
------	--	----------